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Exp-15: Implement Iris Flower Classification using Naive Bayes classifier

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# Import required libraries

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix

# Load the Iris dataset

iris = load_iris()

X = iris.data

y = iris.target

# Split the dataset into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create the Naive Bayes model

model = GaussianNB()

# Train the model

model.fit(X_train, y_train)

# Predict the test data

y_pred = model.predict(X_test)

# Display results

print("Actual Labels:")

print(y_test)

print("\nPredicted Labels:")

print(y_pred)

# Accuracy

accuracy = accuracy_score(y_test, y_pred)
```

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print("\nAccuracy: {:.2f}%".format(accuracy * 100))

# Confusion Matrix

cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")

print(cm)
```

Output:

```
... Actual Labels:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]

Predicted Labels:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]

Accuracy: 100.00%

Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```